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(54) **MULTI-BUFFERED REGISTER FILES WITH SHARED ACCESS CIRCUITS**

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(51) **Int. Cl.**

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G06F 13/16	(2006.01)
G06F 15/78	(2006.01)
G06N 3/04	(2023.01)

(52) **U.S. Cl.**

CPC ***G06F 9/30123*** (2013.01); ***G06F 9/30101*** (2013.01); ***G06F 9/3012*** (2013.01); ***G06F 9/3013*** (2013.01); ***G06F 9/30141*** (2013.01); ***G06F 13/1689*** (2013.01); ***G06F 15/7807*** (2013.01); ***G06N 3/04*** (2013.01)

(58) **Field of Classification Search**

CPC .. G06F 9/3012; G06F 9/30123; G06F 9/3013; G06F 9/30141; G06F 15/7807

See application file for complete search history.

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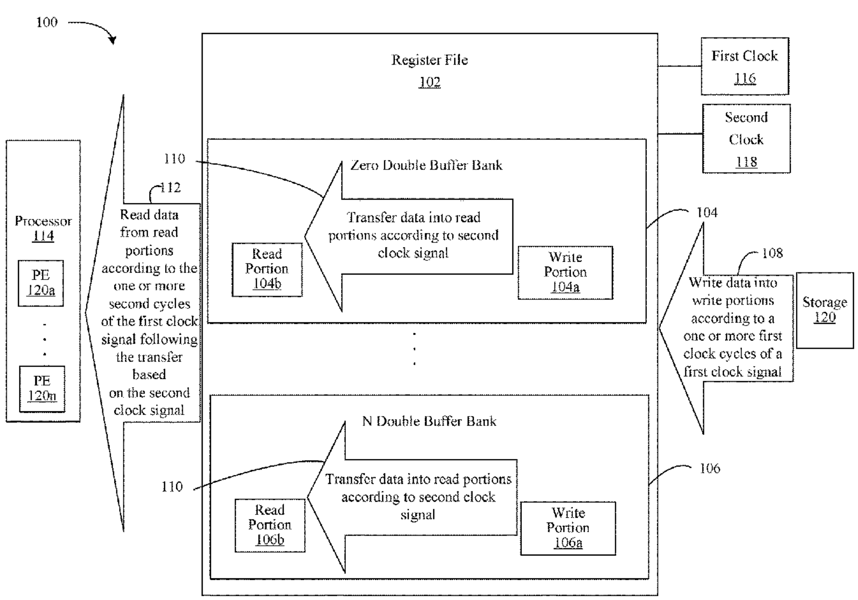
(74) *Attorney, Agent, or Firm* — Akona IP PC

(57) **ABSTRACT**

Systems, apparatuses and methods identify a plurality of registers that are associated with a system-on-chip. The plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations. The technology writes data to the first portion of the plurality of registers, and transfers the data from the first portion to the second portion.

22 Claims, 8 Drawing Sheets



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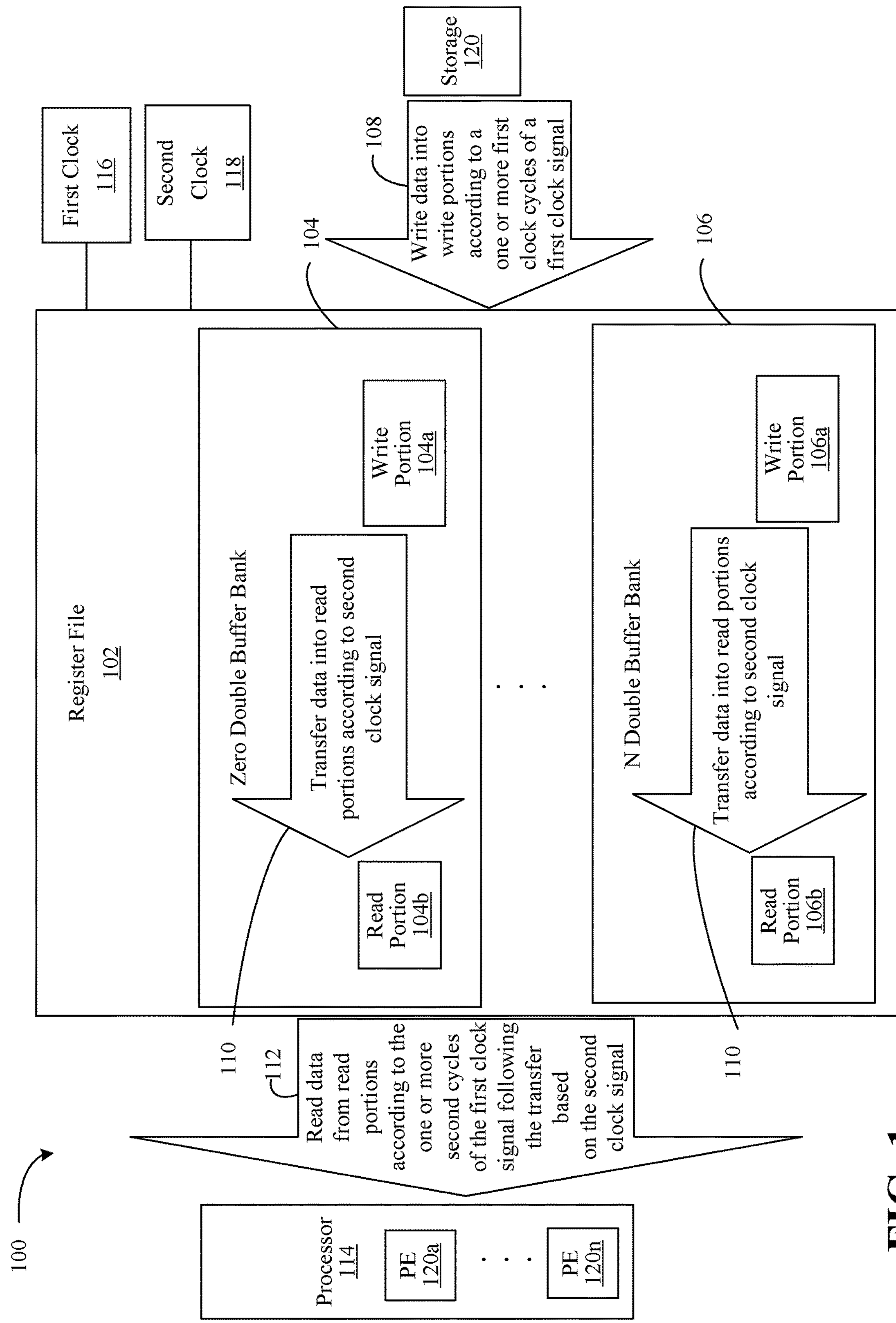

100
First Clock
116
Second Clock
118
Storage
120
108
Write data into write portions according to a one or more first clock cycles of a first clock signal
104
106
Register File
102
Zero Double Buffer Bank
Write Portion
104a
Transfer data into read portions according to second clock signal
Read Portion
104b
N Double Buffer Bank
Write Portion
106a
Transfer data into read portions according to second clock signal
Read Portion
106b
110
110
112
Read data from read portions according to the one or more second cycles of the first clock signal following the transfer based on the second clock signal
Processor
114
PE
120a
PE
120n

FIG. 1

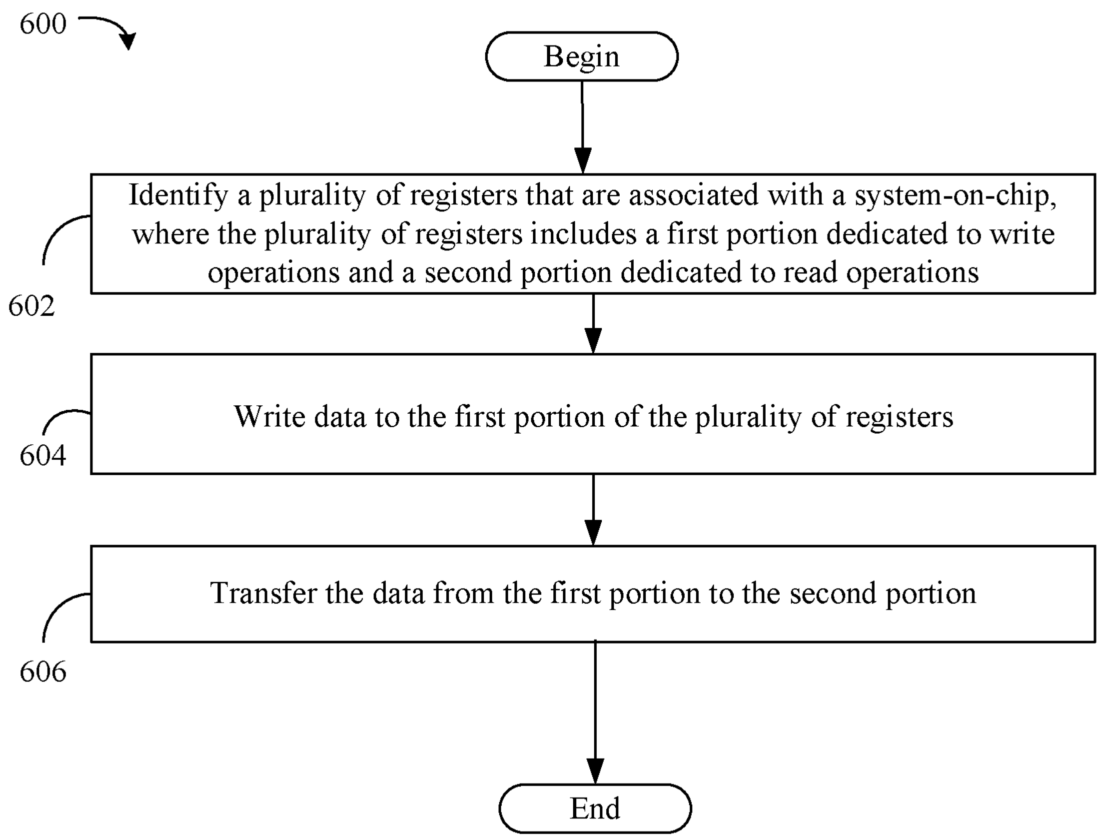

600
Begin
Identify a plurality of registers that are associated with a system-on-chip, where the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations
602
Write data to the first portion of the plurality of registers
604
Transfer the data from the first portion to the second portion
606
End

FIG. 2

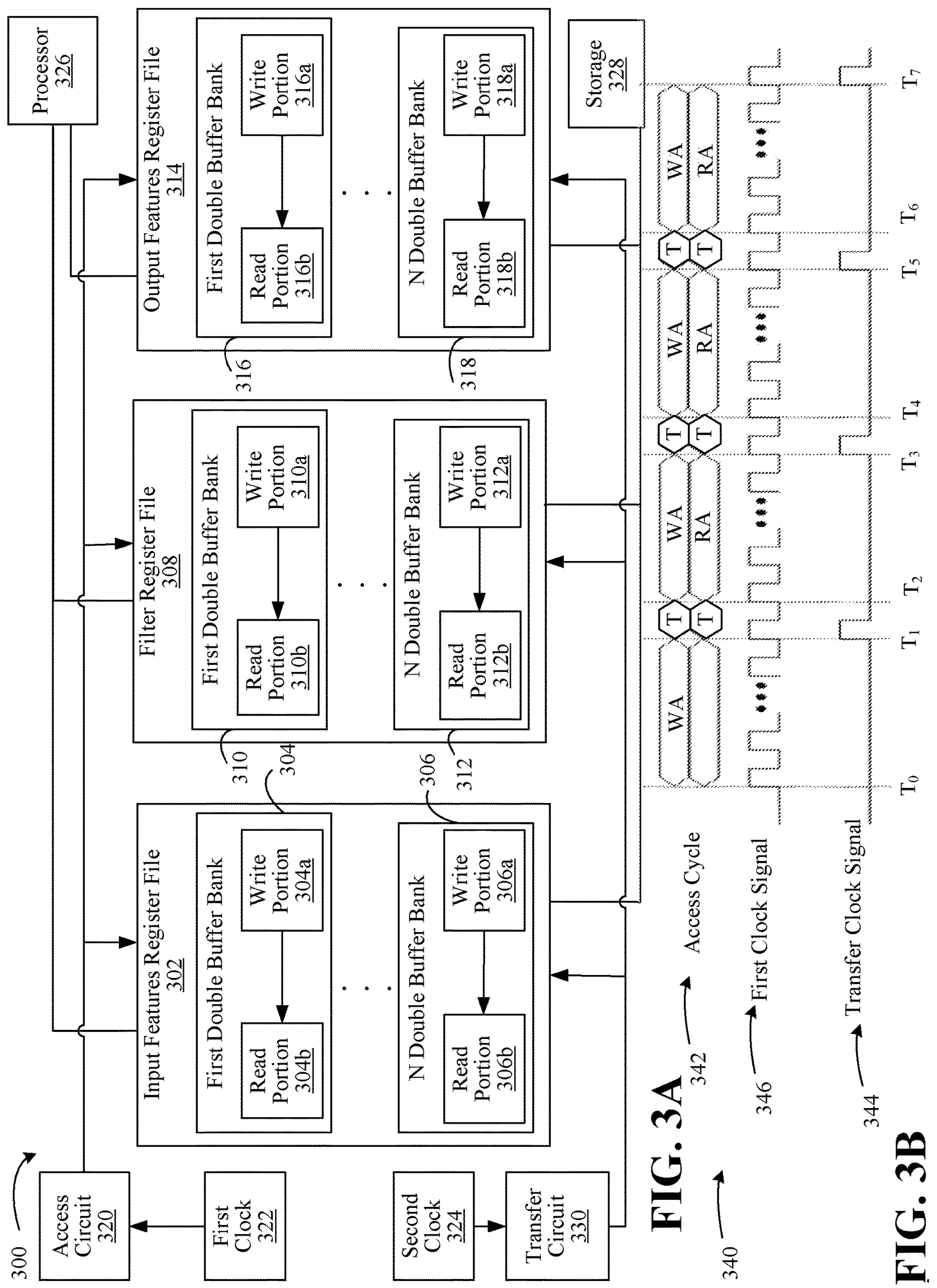
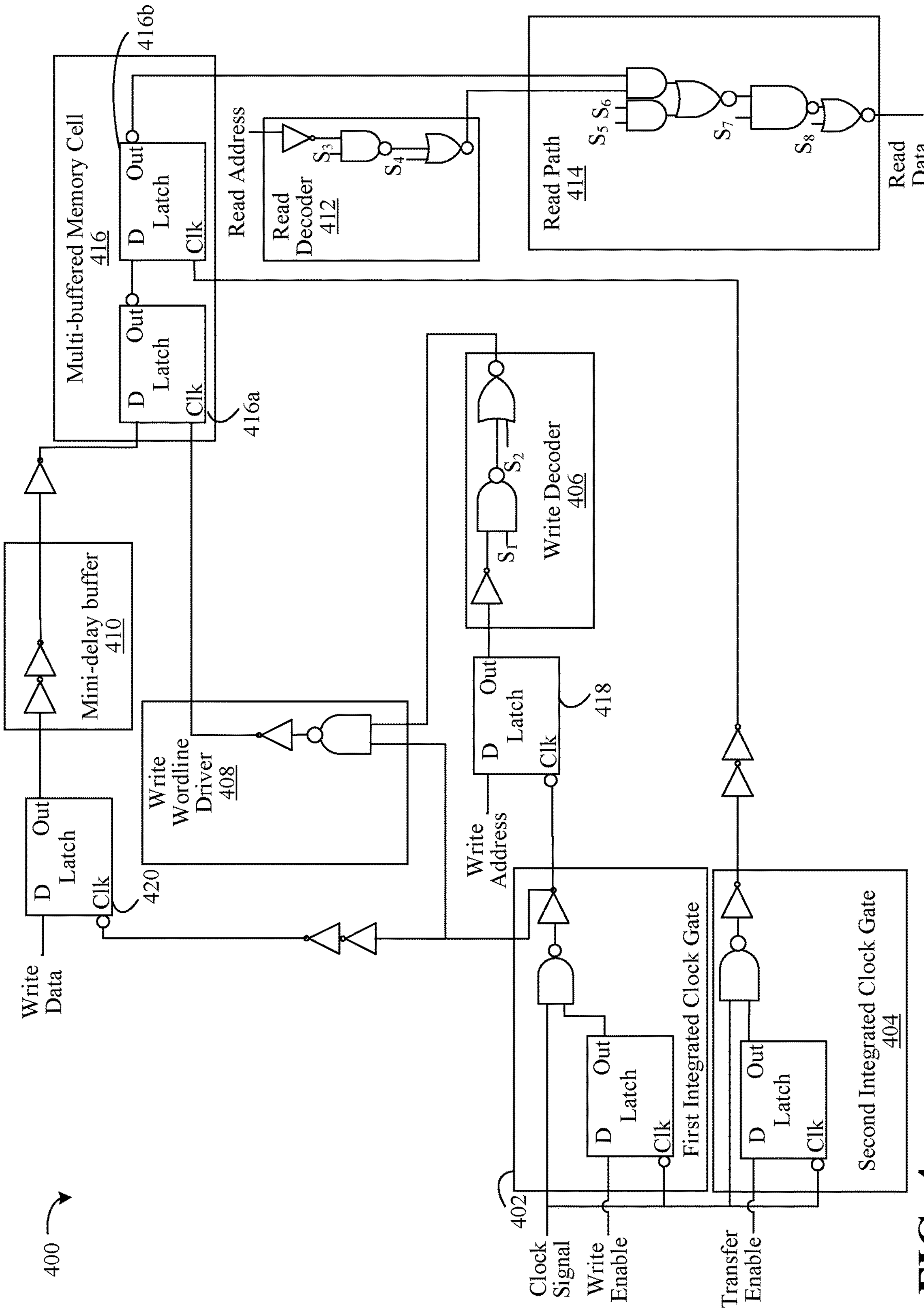

300
Access Circuit 320
First Clock 322
Second Clock 324
Transfer Circuit 330
Processor 326
Storage 328
Input Features Register File 302
First Double Buffer Bank
Write Portion 304a
Read Portion 304b
N Double Buffer Bank
Write Portion 306a
Read Portion 306b
304
306
Filter Register File 308
First Double Buffer Bank
Write Portion 310a
Read Portion 310b
N Double Buffer Bank
Write Portion 312a
Read Portion 312b
310
312
Output Features Register File 314
First Double Buffer Bank
Write Portion 316a
Read Portion 316b
N Double Buffer Bank
Write Portion 318a
Read Portion 318b
316
318
340
342 Access Cycle
346 First Clock Signal
344 Transfer Clock Signal
WA
RA
T
T_0 T_1 T_2 T_3 T_4 T_5 T_6 T_7

FIG. 3A

FIG. 3B



400
Write Data
D Latch Out Clk
420
Mini-delay buffer 410
Multi-buffered Memory Cell 416
D Latch Out Clk
416a
416b
Write Wordline Driver 408
Write Address
D Latch Out Clk
418
Write Decoder 406
S1
S2
Read Address
Read Decoder 412
S3
S4
Read Path 414
S5 S6
S7
S8
Read Data
402
Clock Signal
Write Enable
First Integrated Clock Gate
Transfer Enable
Second Integrated Clock Gate 404

FIG. 4

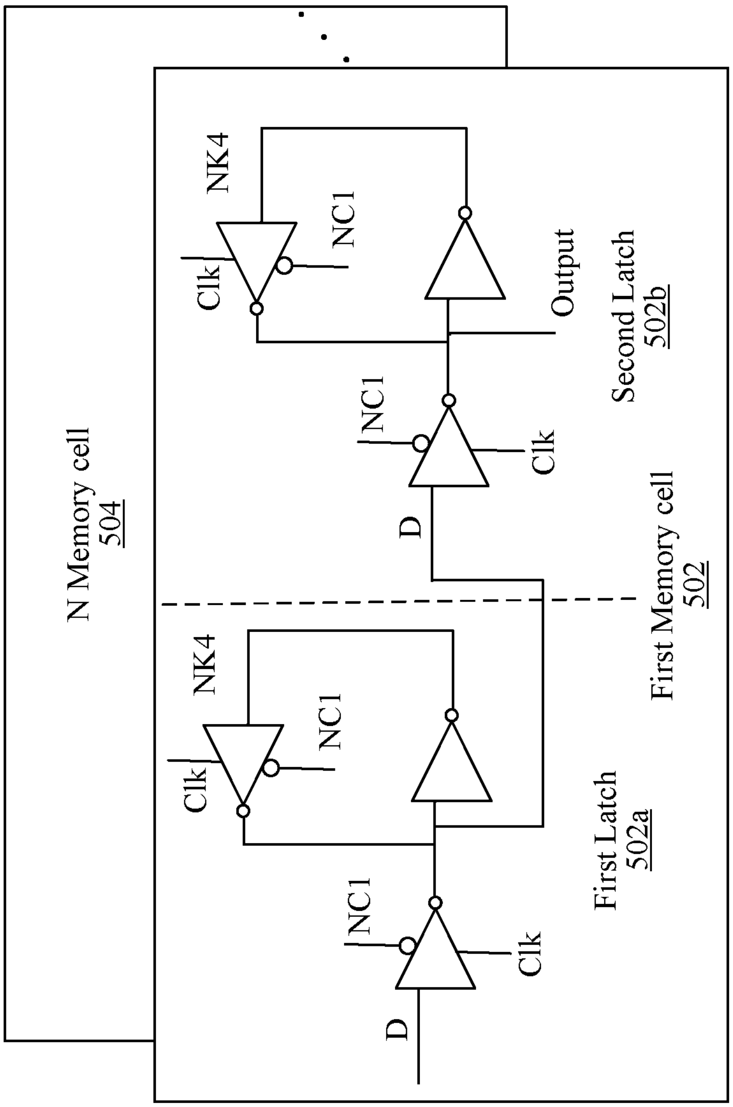

N Memory cell
504
First Memory cell
502
First Latch
502a
Second Latch
502b
D
NC1
Clk
NK4
NC1
Clk
D
NC1
Clk
NK4
NC1
Clk
Output

FIG. 5

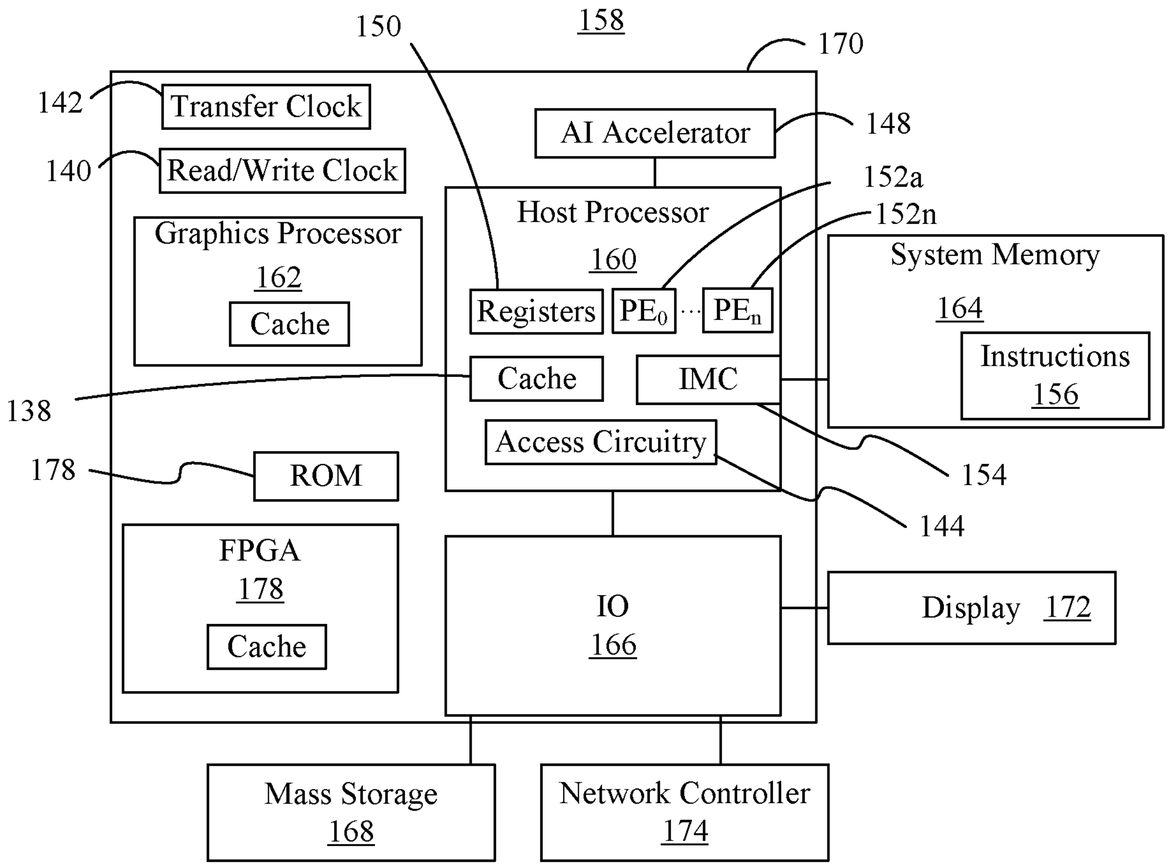

158
150
170
142
Transfer Clock
140
Read/Write Clock
AI Accelerator
148
152a
152n
Host Processor
160
Graphics Processor
162
Cache
Registers
PE0
PEn
System Memory
164
Instructions
156
Cache
IMC
138
Access Circuitry
178
ROM
154
144
FPGA
178
Cache
IO
166
Display 172
Mass Storage
168
Network Controller
174

FIG. 6

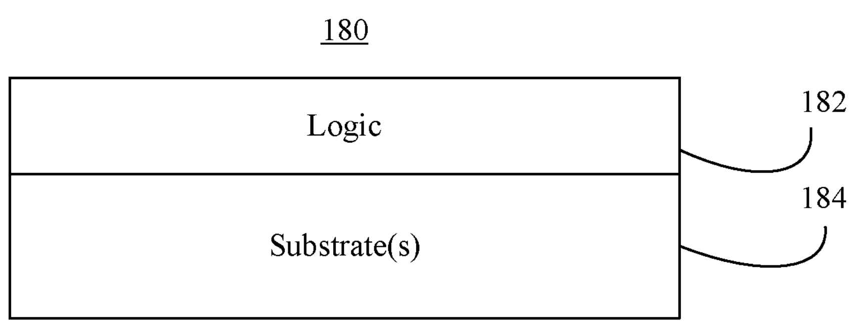

180
Logic
182
184
Substrate(s)

FIG. 7

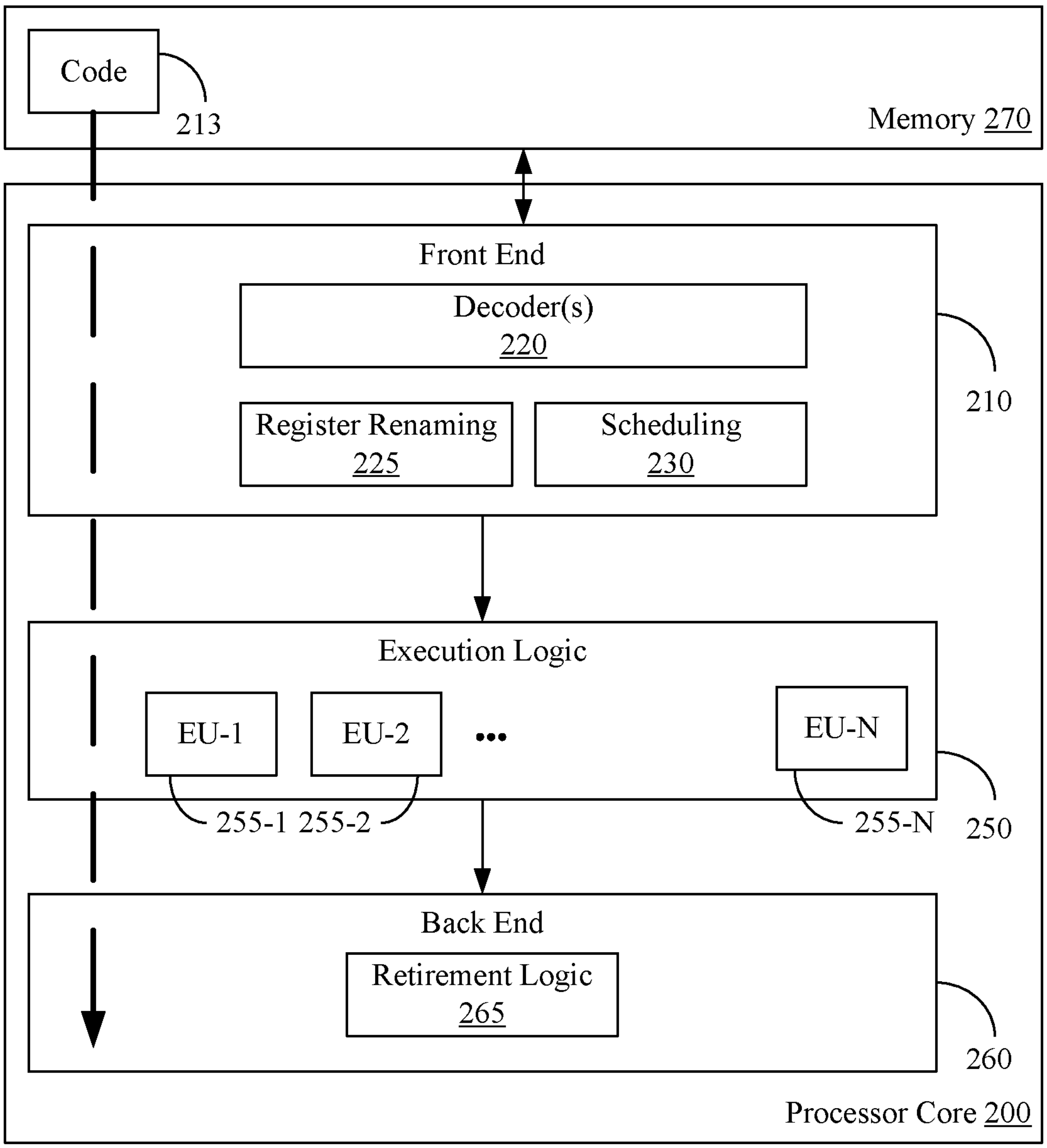
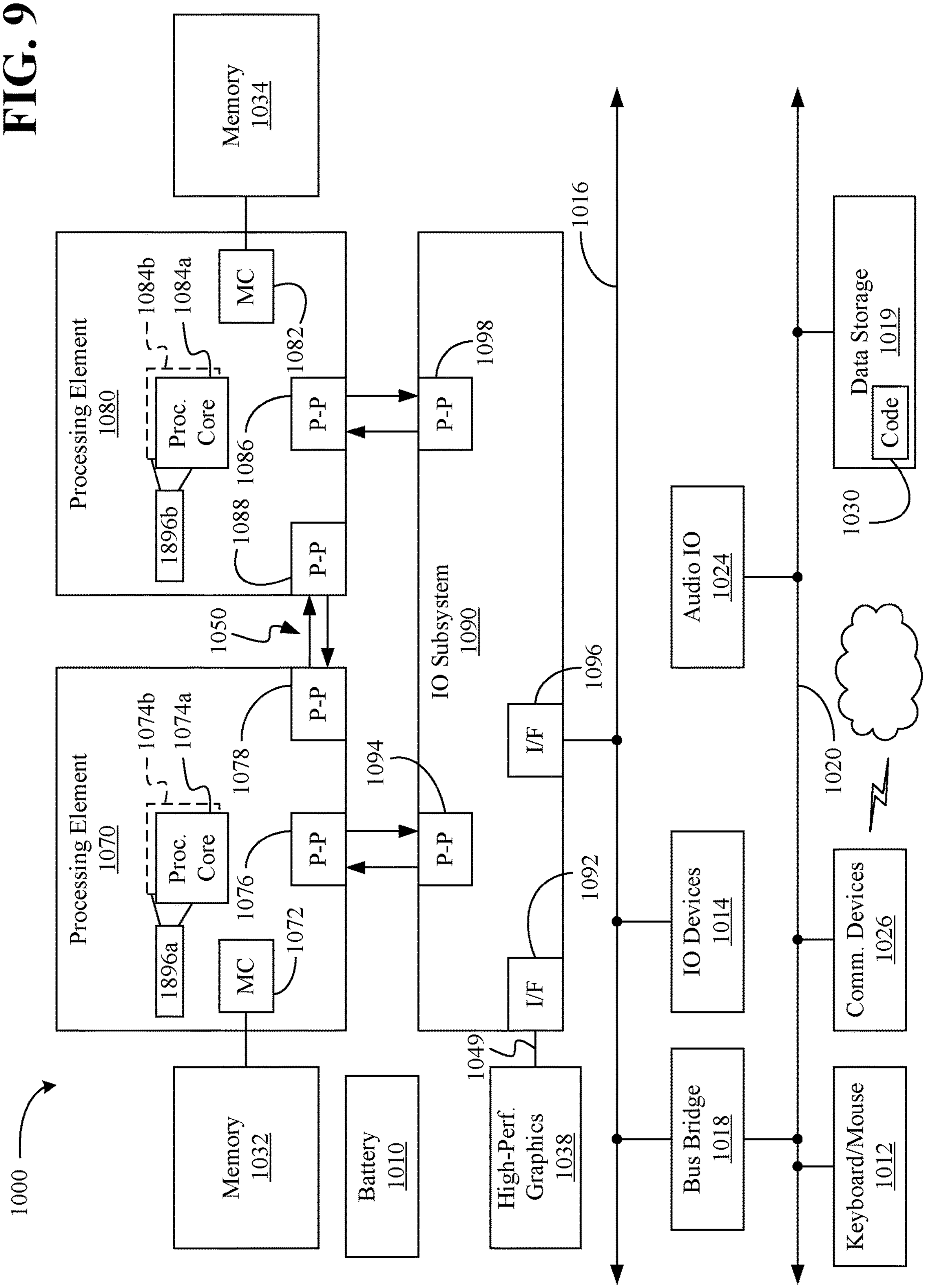

Code
213
Memory 270
Front End
Decoder(s)
220
Register Renaming
225
Scheduling
230
210
Execution Logic
EU-1
EU-2
•••
EU-N
255-1 255-2
255-N
250
Back End
Retirement Logic
265
260
Processor Core 200

FIG. 8

FIG. 9


1000
Memory
1034
Memory
1032
Battery
1010
Processing Element
1080
Processing Element
1070
Proc. Core
Proc. Core
1896b
1896a
MC
MC
P-P
P-P
P-P
P-P
P-P
P-P
1084b
1084a
1074b
1074a
1082
1072
1086
1088
1076
1078
1050
1094
1098
IO Subsystem
1090
I/F
I/F
1092
1096
1049
High-Perf. Graphics
1038
1016
IO Devices
1014
Audio IO
1024
Bus Bridge
1018
Data Storage
1019
Code
1030
1020
Comm. Devices
1026
Keyboard/Mouse
1012

MULTI-BUFFERED REGISTER FILES WITH SHARED ACCESS CIRCUITS

TECHNICAL FIELD

Embodiments generally relate to multi-buffered (e.g., double-buffered) memory implementations. More particularly, embodiments relate to shared access circuitry (e.g., read/write access circuitry) associated with the multi-buffered memory implementations.

BACKGROUND

In an attempt to meet workload demands, some examples increase different types of processing units or elements (e.g., multiply-and-accumulate (MAC) units) to increase processing power. The processing units may nonetheless suffer from degraded performance due to blocking caused by high latency data fetch and/or memory access operations. Thus, merely scaling compute power may not yield performance gains. Indeed, including idle processing units may be detrimental to the overall power/energy of the system since idling processing units may increase the leakage component, increase hardware costs and a size of an overall package.

BRIEF DESCRIPTION OF THE DRAWINGS

The various advantages of the embodiments will become apparent to one skilled in the art by reading the following specification and appended claims, and by referencing the following drawings, in which:

FIG. **1** is a process flow diagram of an example of a multi-buffered memory access according to an embodiment;

FIG. **2** is a flowchart of an example of a method of accessing multi-buffered registers according to an embodiment;

FIG. **3**A is a block diagram of an example of a double-buffer register file architecture according to an embodiment;

FIG. **3**B is a timing diagram of controlling the double-buffer register file according to an embodiment;

FIG. **4** is an electrical diagram of an example of a double-buffer register file architecture according to an embodiment;

FIG. **5** is an electrical diagram of an example of memory cells of a double-buffer register file architecture according to an embodiment;

FIG. **6** is a block diagram of an example of a performance-enhanced computing system according to an embodiment;

FIG. **7** is an illustration of an example of a semiconductor apparatus according to an embodiment;

FIG. **8** is a block diagram of an example of a processor according to an embodiment; and

FIG. **9** is a block diagram of an example of a multi-processor based computing system according to an embodiment.

DESCRIPTION OF EMBODIMENTS

Turning now to FIG. **1**, a multi-buffered memory access process **100** is illustrated that provides a flexible schedule based approach of data movements to reduce overall memory circuitry and latency of memory fetches. The process **100** may move data from lower levels of memory, such as storage **120**, to higher levels of a memory hierarchy such as register file **102** (e.g., a hardware register) having dedicated write portions **104***a*, **106***a* and read portions **104***b*, **106***b*. The register file **102** may include a number of data banks, including the zero double buffer bank **104** and the N double buffer bank **106**. The register file **102** may include shared read-write circuitry to store data and reduce blocking of the processor **114** due to memory accesses, while also reducing an overall size of the register file **102**.

The process **100** may include writing data from storage **120** into write portions **104***a*, **106***a* according to one or more first clock cycles of a first clock signal **108** of the first clock **116**. The register file **102** may be a local storage (e.g., memory device) of a processor **114** (e.g., on-chip registers) and may be allocated close to compute processing elements **120***a*-**120***n*. The register file **102** may be accessed more frequently than distant, higher levels of memory hierarchy such as storage **120** (e.g., compute-near-Memory) which may be a static random-access memory (SRAM) and/or dynamic random access memory (DRAM). Thus, some embodiments include register files **102** that are close to processing elements **120***a*-**120***n* to enable reuse of data to bypass higher latency memory fetches.

The process **100** may include transferring data into read portions **104***b*, **106***b* according to a second clock signal (e.g., a pulse) of the second clock **118**, **110**. Each clock cycle of the second clock signal may correspond (e.g., may be as long as) several cycles of the first clock signal. In some embodiments, the second clock **118** may be omitted and the second clock signal may be a modified version of the first clock signal. The second clock signal is different from the first clock signal to avoid overwriting data in the read portions **104***b*, **106***b* that is not yet read by the processor **114**. After the data in the read portions **104***b*, **106***b* is read by the processor **114**, new data may be transferred (e.g., according to the second clock signal) from the write portions **104***a*, **106***a* to overwrite the data in the read portions **104***b*, **106***b* and be stored in the read portions **104***b*, **106***b*. In some embodiments, the second clock signal is timed to cause transfer of data from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* between read operations of the read portions **104***b*, **106***b* by the processor **114**, and write operations of data from storage **120** to the write portion **104***a*, **106***a*. The data may be transferred from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* in one cycle of the second clock signal.

Process **100** may further read data from read portions **104***b*, **106***b*, according to one or more second clock cycles of the first clock signal following (e.g., immediately subsequent to) the transfer based on the second clock signal **112**. For example, the processor **114** may read the data stored in read portions **104***b*, **106***b* to execute operations based on the data with the PE **120***a*-**120***n*. In some embodiments, the processor **114** may only be able to access the read portions **104***b*, **106***b* and not the write portions **104***a*, **106***a*. Further, data from the storage **120** may only be directly written into the write portions **104***a*, **106***a* and not the read portions **104***b*, **106***b*. Doing so may be reduce the circuitry for accessing the zero double buffer bank **104** and the N double buffer bank **106** since duplicative circuitry may be eliminated. Thus, the write portions **104***a*, **106***a* may be dedicated to write operations from the storage **120**, while the read portion **104***b*, **106***b* may be dedicated to read operations from the processor **114**.

For example, the write portions **104***a*, **106***a* may receive data that is external to the register file **102** and from elements external to the register file **102**. The write portions **104***a*, **106***a* may not be read by elements external to the register file **102**. The read portions **104***b*, **106***b* may be read by elements external to the register file **102**, but may not be written to by elements external to the register file **102**. Data may be transferred internally within the register file **102** from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b*, but as noted, external elements may have limited interactions with the write portions **104***a*, **106***a* to bypass reading the write portions **104***a*, **106***a*, and bypass writing to the read portions **104***b*, **106***b*.

During consecutive clock cycles of the first clock signal of first clock **116**, data may be written into the write portions **104***a*, **106***a*, and read from read portions **104***b*, **106***b*. Data may be transferred from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* between portions of the clock cycles (e.g., between phases) of the first clock **116** to avoid data errors such as overwriting. For example, data may be written into the write portions **104***a*, **106***a* during one or more first clock cycles of the first clock signal. During subsequent clock cycles, the loaded data may be transferred from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* based on the second clock signal. During one or more subsequent clock cycles of the first clock signal after the second clock signal has transferred data from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b*, the processor **114** may read data from the read portions **104***b*, **106***b*.

Such operations may repeat over the first and second clock signals. For example, in one or more first clock cycles of the first clock signal, first data may be read from the read portions **104***b*, **106***b* (that may have been stored in one or more previous clock cycles of the first clock signal) and second data may be stored in the write portions **104***a*, **106***a* from the storage **120**. The second data may be transferred from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* according to the second clock signal and to overwrite the first data after the first data is read from the read portions **104***b*, **106***b*. In some embodiments, the read of the first data from the read portions **104***b*, **106***b* and write of the second data in the write portions **104***a*, **106***a* may occur concurrently to avoid overwriting errors.

Thereafter, in one or more second clock cycles of the first clock signal, the second data may be read from the read portions **104***b*, **106***b* (that were stored in the one or more first clock cycles) and third data may be stored in the write portions **104***a*, **106***a*. The third data may be transferred from the write portions **104***a*, **106***a* to the read portions **104***b*, **106***b* according to the second clock signal and to overwrite the second data. In some embodiments, the read of the second data from the read portions **104***b*, **106***b* and write of the third data may in the write portions **104***a*, **106***a* may occur concurrently. Such a process **100** may continue for a plurality of consecutive clock cycles of the first clock signal and the second clock signal.

By including dedicated write portions **104***a*, **106***a*, and dedicated read portions **104***a*, **104***b*, the register file **102** may be an area-efficient, multi-buffered (e.g., double-buffered) memory with shared read/write circuits to reduce area (e.g., by over 49% or more) over other implementations while reducing waiting by the PEs **120***a*-**120***n* for memory fetches. Thus, some embodiments enhance throughput at a reduced memory size by sharing the read and write circuitry between the first (e.g., active) register file and a buffered (e.g., shadow) register file to reduce memory size.

In some embodiments, more than one register file **102** may be provided. For example, a plurality of register files **102** may be included depending on the application. For example, if a deep neural network is implemented, first register file may be dedicated to input features (IF), a second register file may be dedicated to filters (FL) a third register file may be dedicated to output features (OF) register (e.g., for calculations that are determined based on the FL and the IF). If the register file **102** is modified to be an OF register, the above process **100** may be reversed so that the processor **114** may write to dedicated write portions, data may be transferred from the write portions to read portions and then transferred to another storage from the read portions.

Thus, some embodiments may maintain high throughput performance at a reduced cost and size by avoiding duplicative hardware overheads implemented by other designs (e.g., a ping-pong double-buffered register file implementations). In some embodiments, the PEs **120***a*-**120***n* may include deep neural network (DNN) accelerators and/or arithmetic logic units (ALU), graphics accelerators, artificial intelligence accelerators, etc. Some embodiments may be implemented in conjunction with high performance microprocessors, graphics, and other application specific hardware accelerators.

For example, in a higher performance microprocessor, a double-buffered register file such as register file **102** may be an instance of one set of register files being read by the ALU and another set of register files are written to by the ALU in order to achieve computational overlaps for enhancing performance. The process **100** may be implemented in high performance general purpose computing processes, graphics processes, and/or other hardware accelerator processes. Some embodiments may be applicable to artificial intelligence applications such as imaging, video, and speech recognition.

FIG. **2** shows a method **600** of accessing multi-buffered registers. The method **600** may generally be implemented with the embodiments described herein, for example, the process **100** (FIG. **1**), already discussed. In an embodiment, the method **600** is implemented in one or more modules as a set of logic instructions stored in a machine- or computer-readable storage medium such as random access memory (RAM), read only memory (ROM), programmable ROM (PROM), firmware, flash memory, etc., in configurable logic such as, for example, programmable logic arrays (PLAs), field programmable gate arrays (FPGAs), complex programmable logic devices (CPLDs), in fixed-functionality logic hardware using circuit technology such as, for example, application specific integrated circuit (ASIC), complementary metal oxide semiconductor (CMOS) or transistor-transistor logic (TTL) technology, or any combination thereof.

For example, computer program code to carry out operations shown in the method **600** may be written in any combination of one or more programming languages, including an object oriented programming language such as JAVA, SMALLTALK, C++ or the like and conventional procedural programming languages, such as the "C" programming language or similar programming languages. Additionally, logic instructions might include assembler instructions, instruction set architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, state-setting data, configuration data for integrated circuitry, state information that personalizes electronic circuitry and/or other structural components that are native to hardware (e.g., host processor, central processing unit/CPU, microcontroller, etc.).

Illustrated processing block **602** includes identifying a plurality of registers that are associated with a system-on-chip. The plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations. Illustrated processing block **604** writes data to the first portion of the plurality of registers. Illustrated processing block **606** transfers the data from the first portion to the second portion. In some embodiments, the method **600** includes transferring the data in response to a transfer clock pulse being identified. In some embodiments, the method **600** further includes writing the data to the first portion during one or more first clock cycles prior to the transfer clock pulse being identified. In some embodiments, the method **600** further includes reading the data in the first portion during one or more second clock cycles after the transfer clock pulse is identified.

In some embodiments, the plurality of registers includes a first register associated with input features associated with a neural network, a second register associated with filters associated with the neural network, and a third register associated with output features associated with the neural network. In some embodiments, the method **600** further includes in each of a plurality of consecutive clock cycles, retrieving data from a memory and storing the retrieved data in the first portion. Furthermore, some embodiments of the method **600** include in each of a plurality of consecutive clock cycles, reading data from the second portion. In some embodiments, the method **600** further includes controlling read and write operations to the first and second portions based on a first clock signal and transferring data between the first and second portions based on a second clock signal that is different from the first clock signal.

FIG. **3**A illustrates a double-buffer register file architecture **300** with shared read and write circuitry. FIG. **3**B illustrates timing diagrams **340** of accesses and transfers of data in the architecture **300**. Architecture **300** may be associated with a machine learning model such as a DNN and/or CNN. First clock **322** may generate the first clock signal **346** while a second clock **324** may generate the transfer clock signal **344**.

As illustrated in FIG. **3**A, the architecture **300** includes an input features register file **302**, filter register file **308** and output features register file **314**. The input features register file **302** includes first double buffer bank **304**-N double buffer bank **306**. The write portions **304***a*, **306***a* may each include one write port in some embodiments, and the read portions **304***b*, **306***b* may each include a plurality of read ports (e.g., 4 or more) in some embodiments.

The filter register file **308** includes first double buffer bank **310**-N double buffer bank **312**. Write portions **310***a*, **312***a* may each include one write port in some embodiments, and the read portions **310***b*, **312***b* may each include one read port in some embodiments.

The output features register file **314** includes first double buffer bank **316**-N double buffer bank **318**. Write portions **316***a*, **318***a* may each include one write port in some embodiments, and the read portions **316***b*, **318***b* may each include one read port in some embodiments.

Access circuit **320** may control write accesses to the input features register file **302**, filter register file **308** and output features register file **314**. In some embodiments, the access circuit **320** may be a part of the input features register file **302**, filter register file **308** and output features register file **314**. Furthermore, while the access circuit **320** is shown schematically as one component, it will be understood that other implementations may be possible. For example, each of the input features register file **302**, filter register file **308** and output features register file **314** may include different logic that controls write operations and read operations as described herein and similar to the access circuit **320**. The access circuit **320** may include hardware, such as configurable logic and/or fixed-functionality hardware logic.

Access circuit **320** may store data from storage **328** (e.g., solid state drive, hard disk drive, SRAM, DRAM, etc.) into the write portions **304***a*, **306***a*, **310***a*, **312***a* and in accordance with a first clock signal **346** (FIG. **3**B) from the first clock **322**. The access circuit **320** may store data computed by processor **326** into the write portions **316***a*, **318***a* and in accordance with the first clock signal **346**. The first clock signal **346** may correspond to access cycles **342** of timing diagram **340** (FIG. **3**B). The upper graph of access cycles **342** corresponds to actions (e.g., writes into) associated with the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* and the lower graph of access cycles **342** corresponds to actions (e.g., reads from) associated with the read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b*. Further, transfers may occur between read and writes. In the example of access cycles **342**, the write actives are denoted with a "WA," the read actives are denoted with a "RA," and transfers are denoted with a "T." Multiple reads and writes may occur during each read active and write active respectively. Further, multiple transfers may occur during each transfer. The access circuit **320** may include configurable logic and/or fixed-functionality logic hardware.

Further, the access circuit **320** may control read accesses to the input features register file **302**, filter register file **308** and output features register file **314**. For example, access circuit **320** may read data out of the read portions **304***b*, **304***b*, **310***b*, **312***b* in accordance with the first clock signal **346** from the first clock **322** to provide the data to the processor **326**. The access circuit **320** may read data computed by processor **326** and stored in the read portions **316***b*, **318***b* to store the data to the storage **328** and in accordance with the first clock signal **346**.

In some embodiments and as described above, the transfer circuit **330** may transfer data from the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* to the read portions **304***b*, **304***b*, **310***b*, **312***b*, **316***b*, **318***b* in accordance with a transfer clock signal of the second clock **324** during transfer cycles. The transfer clock signal **344** may correspond to transfer cycles of timing diagram **340** (FIG. **3**B).

Some embodiments may replace duplicative read-write circuitry that may otherwise be present in each of the two instances of a double-buffer register file with a single write and single read circuitry to control accesses. For instance, the access circuit **320** may include a single write and single read circuitry that is common to the first double buffer bank **304** to access read portion **304***b* and write portion **304***a*. Notably, at least part of the write circuitry may not be needed for the read portion **304***b* since components external to the input features register file **302** do not write to the read portion **304***b*. Rather, data is transferred internally from the write portion **304***a* to the read portion **304***b*. Further, read circuitry may not be needed for the write portion **304***a* since the write portion **304***a* is not read by components external to the input features register file **302**. Similarly, the access circuit **320** may include a single write and single read circuitry that is common to the N double buffer bank **306** to access the read portion **306***b* and write portion **306***a*. The access circuit **320** may similarly include single read write circuitry for the first-N double buffer banks **310**-**312** and the first-N double buffer banks **316**-**318**.

In some embodiments, the write circuitry of the access circuit **320** only accesses the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* (e.g., active register file bit-cells). Furthermore, in some embodiments the read circuitry of the access circuit **320** is only allowed to access read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b* (e.g., shadow register file bit-cell).

For example, in access cycles **342**, during first clock cycles from T_0-T_1, first data may be written into write portions **304***a*, **306***a*, **310***a*, **312***a* from storage **328**. Since this is the first clock cycles, the read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b* do not yet contain data, and thus no data is yet passed to the processor **326**. After the first data has been loaded into the write portions **304***a*, **306***a*, **310***a*, **312***a*, a transfer pulse of transfer clock signal **344** may trigger a transfer from time T_1-T_2 to move the first data from the write portions **304***a*, **306***a*, **310***a*, **312***a* to the read portions **304***b*, **306***b*, **310***b*, **312***b*. In some embodiments, time T_1-T_2 may overlap with the first clock cycles by avoiding overwriting data that has not yet been read by the processor **326** or transferred to storage **328**.

Thereafter, from time T_2-T_3 during second clock cycles of the first clock signal **346** from time T_2-T_3, the first data may be read from the read portions **304***b*, **306***b*, **310***b*, **312***b* by the processor **326**. Furthermore, from time T_2-T_3, second data may be written into the write portions **304***a*, **306***a*, **310***a*, **312***a* from storage **328**. The processor **326** may now being to execute operations based on the first data. After the second data has been loaded into the write portions **304***a*, **306***a*, **310***a*, **312***a* and the processor **326** has read the first data from the read portions **304***b*, **306***b*, **310***b*, **312***b*, a transfer pulse may trigger a transfer from time T_3-T_4 to move the second data from the write portions **304***a*, **306***a*, **310***a*, **312***a* to the read portions **304***b*, **306***b*, **310***b*, **312***b*. In some embodiments, time T_3-T_4 may overlap with the second clock cycles by avoiding overwriting data that has not yet been read by the processor **326** or transferred to storage **328**.

Thereafter, from time T_4-T_5 during third clock cycles of the first clock signal **346**, the second data may be read from the read portions **304***b*, **306***b*, **310***b*, **312***b* by the processor **326**. Furthermore, from time T_4-T_5, third data may be written into the write portions **304***a*, **306***a*, **310***a*, **312***a* from storage **328**. The processor **326** may have completed operations based on the first data and stores the output in the write portions **316***a*, **318***a* during the third clock cycles. After the third data and output have been loaded into the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* and the processor **326** has read the second data from the from the read portions **304***b*, **306***b*, **310***b*, **312***b*, a transfer pulse may trigger a transfer from time T_5-T_6 to move the third data from the write portions **304***a*, **306***a*, **310***a*, **312***a* to the read portions **304***b*, **306***b*, **310***b*, **312***b*, and the output from the write portion **316***a*, **318***a* to the read portions **316***b*, **318***b*. In some embodiments, the time T_5-T_6 may overlap with the third clock cycles by avoiding overwriting data that has not yet been read by the processor **326** or transferred to storage **328**.

The above process may repeat again starting at time T_6 and similar to the above. For example, the read portions **316***b*, **318***b* may be read to identify the output and store the output in the storage **328**. In some embodiments, the transfer cycles of the transfer clock signal **344** correspond to a clock pulse that triggers the copy of the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* (e.g., active register file bit-cells) to read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b* (e.g., shadow register file bit-cell). The write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* may operate solely in the write phase, while the read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b* operate solely in the read phase with a pulse of a trigger clock, such as transfer clock signal **344**, between the two phases that causes copying of contents from the write portions **304***a*, **306***a*, **310***a*, **312***a*, **316***a*, **318***a* into the read portions **304***b*, **306***b*, **310***b*, **312***b*, **316***b*, **318***b*.

FIG. **4** illustrates an example of a double-buffered register file architecture **400** with shared read/write circuits. As illustrated, a first integrated clock gate **402** may be provided to generate a first clock signal associated with a write portion **416***a* of the multi-buffered memory cell **416**. The write enable signal may be provided to the D input of the latch of the first integrated clock gate **402** and the clock signal may be provided to the clock input of the latch of first integrated clock gate **402**.

A write decoder **406** may receive an input of a write address from a latch **418** at a timing corresponding to the first clock signal. The write decoder **406** may provide an output to the write wordline driver **408**, which provides an output to the clock input of the write portion **416***a*. The write data may be transmitted to the multi-buffered memory cell through a mini-delay buffer **410** and a latch **420**.

A second integrated clock gate **404** may provide a second clock signal to the clock input of a read portion **416***b* of the multi-buffered memory cell **416**. The data may be transferred from the write portion **416***a* to the read portion **416***b* based on a timing of the second clock signal. A read decoder **412** and read path **414** may provide read data from the read portion **416***b*. In some embodiments, more than one multi-buffered memory cell **416** may utilize the read path **414**, and hence more than one AND logic gate may be provided in the read path **414**. It is to be noted that the illustrated logic gates show examples of exemplary paths for the various illustrated blocks, such as write decoder **406**, read decoder **412** and read path **414**. In some embodiments the logic pins associated with signals S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8 may be connected to other gate outputs (e.g., other memory cells) which are not shown for clarity to simplify the discussion. The other gate outputs may generate signals S_1, S_2, S_3, S_4, S_5, S_6, S_7, S_8.

For example, the logic pins in the first AND-OR-Invert (AOI) gate in the read path **414** may be connected to two memory cells, and/or two read wordlines (from read decoder) that produce signals S_5, S_6. The next NAND gate may receive inputs from two AOI gates, one of which is unillustrated and generates signal S_7. The last NOR gate may receive inputs from two NAND gates, one of which is unillustrated and produced signal S_8. If the register file has more entries, the read decoder **412** and read path **414** logic may be modified to increase logic depth.

Furthermore, in some embodiments an exposed state node may need to use relative placement so that the exposed state node may be placed next to the first AOI gate of the read path **414**, which corresponds to one or more of a plurality of the multi-buffered memory cells **416**, may utilize the read path **414**. The read decoder **412** may feed into each AND gate of the read path **414** to control the output of the correct multi-buffered memory cell of the multi-buffered memory cells **416**.

In some embodiments, some of the components of FIG. **4** may be duplicated. Some embodiments of FIG. **4** include a double-buffered, quad-latch based memory circuit as the multi-buffered memory cell **416**, where the double-buffered memory cell contains two latches that correspond to read portion **416***b* and write portion **416***a* (e.g., a shadow latch for writing and an active latch for reading). Double-buffered memories may be constructed using quad-latches exposed state node quad latch standard cells.

FIG. **5** illustrates memory cell **502**-N memory cell **504** that may correspond to double-buffered latches. For example, if the memory cells **502**-N memory cells **504** are double buffered quad bit latches, there may be four memory cells total (e.g., N=4). As illustrated a first latch **502***a* may couple with a second latch **502***b*. The memory cells **502**-N memory cells **504** may be readily substituted for the multi-buffered memory cell **416** (FIG. **4**).

Turning now to FIG. **6**, a performance-enhanced double-buffered memory computing system **158** is shown. The system **158** may generally be part of an electronic device/platform having computing functionality (e.g., personal digital assistant/PDA, notebook computer, tablet computer, convertible tablet, server), communications functionality (e.g., smart phone), imaging functionality (e.g., camera, camcorder), media playing functionality (e.g., smart television/TV), wearable functionality (e.g., watch, eyewear, headwear, footwear, jewelry), vehicular functionality (e.g., car, truck, motorcycle), robotic functionality (e.g., autonomous robot), etc., or any combination thereof. In the illustrated example, the system **158** includes a host processor **160** (e.g., CPU) having an integrated memory controller (IMC) **154** that is coupled to a system memory **164**. The host processor **160** includes processing elements including PE_0-PE_n **152***a*-**152***n* that may execute operations based on data on the registers **150** (e.g., hardware registers). The registers **150** may implement a multi-buffer architecture as described herein to include read portions dedicated to providing data to PE_0-PE_n **152***a*-**152***n*, FPGA **178**, graphics processor **162** and/or AI accelerator **148**. The registers **150** may further include write portions to store data from one or more of a cache **138**, system memory **164**, cache of the graphics processor **162** or cache of the FPGA **178**. The access circuitry **144** may control read and writes to the registers **150** in accordance with a clock signal from a read/write clock **140**. The registers **150** may move data from the write portions to the read portions in accordance with a transfer clock signal of transfer clock **142** as described herein.

The illustrated system **158** also includes an input output (IO) module **166** implemented together with the host processor **160** and a graphics processor **162** (e.g., GPU) on a semiconductor die **170** as a system on chip (SoC). The illustrated IO module **166** communicates with, for example, a display **172** (e.g., touch screen, liquid crystal display/LCD, light emitting diode/LED display), a network controller **174** (e.g., wired and/or wireless), and mass storage **168** (e.g., hard disk drive/HDD, optical disk, solid state drive/SSD, flash memory). Furthermore, the SoC **170** may further include processors (not shown) and/or AI accelerator **148** dedicated to artificial intelligence (AI) and/or neural network (NN) processing. For example, the system SoC **170** may include vision processing units (VPUs) and/or other AI/NN-specific processors such as AI accelerator **148**, etc. In some embodiments, any aspect of the embodiments described herein may be implemented in the processors and/or accelerators dedicated to AI and/or NN processing such as AI accelerator **148**, the graphics processor **162** and/or the host processor **160**.

The host processor **160**, the graphics processor **162**, the FPGA **178** and/or the IO module **166** may execute instructions **156** retrieved from the system memory **164** and/or the mass storage. When the instructions **156** are executed, the computing system **158** may implement one or more aspects of the embodiments described herein. For example, the system **158** may implement and/or include one or more aspects of the process **100** (FIG. **1**), the method **600** (FIG. **2**), the architecture **300** (FIG. **3**), the architecture **400** (FIG. **4**), the memory cells **502-504** (FIG. **5**), already discussed.

The illustrated computing system **158** is therefore considered to be performance-enhanced at least to the extent that it enables the computing system **158** to take advantage of low latency memory accesses and storage at reduced footprint and hardware cost. Thus, some embodiments may increase throughput while also reducing a size of the package.

FIG. **7** shows a semiconductor apparatus **180** (e.g., chip, die, package). The illustrated apparatus **180** includes one or more substrates **184** (e.g., silicon, sapphire, gallium arsenide) and logic **182** (e.g., transistor array and other integrated circuit/IC components) coupled to the substrate(s) **184**. In an embodiment, the apparatus **180** is operated in an application development stage and the logic **182** implements and/or includes one or more aspects of the process **100** (FIG. **1**), the method **600** (FIG. **2**), the architecture **300** (FIG. **3**), the architecture **400** (FIG. **4**), the memory cells **502-504** (FIG. **5**), already discussed. Thus, the logic **182** may store buckets that represent a plurality of clusters in a cache, where each of the buckets is to represent a group of the plurality of clusters and further where the plurality of clusters is in a first data format, modify input data from a second data format to the first data format and conduct a similarity search based on the input data in the first data format to assign the input data to at least one bucket of the buckets. Furthermore, the logic **182** may further include processors (not shown) and/or AI accelerator dedicated to artificial intelligence AI and/or NN processing. For example, the system logic **182** may include VPUs, and/or other AI/NN-specific processors such as AI accelerators, etc. In some embodiments, any aspect of the embodiments described herein may be implemented in the processors and/or accelerators dedicated to AI and/or NN processing such as AI accelerators.

The logic **182** may be implemented at least partly in configurable logic or fixed-functionality hardware logic. In one example, the logic **182** includes transistor channel regions that are positioned (e.g., embedded) within the substrate(s) **184**. Thus, the interface between the logic **182** and the substrate(s) **184** may not be an abrupt junction. The logic **182** may also be considered to include an epitaxial layer that is grown on an initial wafer of the substrate(s) **184**.

FIG. **8** illustrates a processor core **200** according to one embodiment. The processor core **200** may be the core for any type of processor, such as a micro-processor, an embedded processor, a digital signal processor (DSP), a network processor, or other device to execute code. Although only one processor core **200** is illustrated in FIG. **8**, a processing element may alternatively include more than one of the processor core **200** illustrated in FIG. **8**. The processor core **200** may be a single-threaded core or, for at least one embodiment, the processor core **200** may be multithreaded in that it may include more than one hardware thread context (or "logical processor") per core.

FIG. **8** also illustrates a memory **270** coupled to the processor core **200**. The memory **270** may be any of a wide variety of memories (including various layers of memory hierarchy) as are known or otherwise available to those of skill in the art. The memory **270** may include one or more code **213** instruction(s) to be executed by the processor core **200**, wherein the code **213** may implement and/or include one or more aspects of the process **100** (FIG. **1**), the method **600** (FIG. **2**), the architecture **300** (FIG. **3**), the architecture **400** (FIG. **4**), the memory cells **502-504** (FIG. **5**), already discussed. The processor core **200** follows a program sequence of instructions indicated by the code **213**. Each instruction may enter a front end portion **210** and be processed by one or more decoders **220**. The decoder **220** may generate as its output a micro operation such as a fixed width micro operation in a predefined format, or may generate other instructions, microinstructions, or control signals which reflect the original code instruction. The illustrated front end portion **210** also includes register renaming logic **225** and scheduling logic **230**, which generally allocate resources and queue the operation corresponding to the convert instruction for execution.

The processor core **200** is shown including execution logic **250** having a set of execution units **255-1** through **255**-N. Some embodiments may include a number of execution units dedicated to specific functions or sets of functions. Other embodiments may include only one execution unit or one execution unit that can perform a particular function. The illustrated execution logic **250** performs the operations specified by code instructions.

After completion of execution of the operations specified by the code instructions, back end logic **260** retires the instructions of the code **213**. In one embodiment, the processor core **200** allows out of order execution but requires in order retirement of instructions. Retirement logic **265** may take a variety of forms as known to those of skill in the art (e.g., re-order buffers or the like). In this manner, the processor core **200** is transformed during execution of the code **213**, at least in terms of the output generated by the decoder, the hardware registers and tables utilized by the register renaming logic **225**, and any registers (not shown) modified by the execution logic **250**.

Although not illustrated in FIG. **8**, a processing element may include other elements on chip with the processor core **200**. For example, a processing element may include memory control logic along with the processor core **200**. The processing element may include I/O control logic and/or may include I/O control logic integrated with memory control logic. The processing element may also include one or more caches.

Referring now to FIG. **9**, shown is a block diagram of a computing system **1000** embodiment in accordance with an embodiment. Shown in FIG. **9** is a multiprocessor system **1000** that includes a first processing element **1070** and a second processing element **1080**. While two processing elements **1070** and **1080** are shown, it is to be understood that an embodiment of the system **1000** may also include only one such processing element.

The system **1000** is illustrated as a point-to-point interconnect system, wherein the first processing element **1070** and the second processing element **1080** are coupled via a point-to-point interconnect **1050**. It should be understood that any or all of the interconnects illustrated in FIG. **9** may be implemented as a multi-drop bus rather than point-to-point interconnect.

As shown in FIG. **9**, each of processing elements **1070** and **1080** may be multicore processors, including first and second processor cores (i.e., processor cores **1074***a* and **1074***b* and processor cores **1084***a* and **1084***b*). Such cores **1074***a*, **1074***b*, **1084***a*, **1084***b* may be configured to execute instruction code in a manner similar to that discussed above in connection with FIG. **8**.

Each processing element **1070**, **1080** may include at least one shared cache **1896***a*, **1896***b*. The shared cache **1896***a*, **1896***b* may store data (e.g., instructions) that are utilized by one or more components of the processor, such as the cores **1074***a*, **1074***b* and **1084***a*, **1084***b*, respectively. For example, the shared cache **1896***a*, **1896***b* may locally cache data stored in a memory **1032**, **1034** for faster access by components of the processor. In one or more embodiments, the shared cache **1896***a*, **1896***b* may include one or more mid-level caches, such as level 2 (L2), level 3 (L3), level 4 (L4), or other levels of cache, a last level cache (LLC), and/or combinations thereof.

While shown with only two processing elements **1070**, **1080**, it is to be understood that the scope of the embodiments are not so limited. In other embodiments, one or more additional processing elements may be present in a given processor. Alternatively, one or more of processing elements **1070**, **1080** may be an element other than a processor, such as an accelerator or a field programmable gate array. For example, additional processing element(s) may include additional processors(s) that are the same as a first processor **1070**, additional processor(s) that are heterogeneous or asymmetric to processor a first processor **1070**, accelerators (such as, e.g., graphics accelerators or digital signal processing (DSP) units), field programmable gate arrays, or any other processing element. There can be a variety of differences between the processing elements **1070**, **1080** in terms of a spectrum of metrics of merit including architectural, micro architectural, thermal, power consumption characteristics, and the like. These differences may effectively manifest themselves as asymmetry and heterogeneity amongst the processing elements **1070**, **1080**. For at least one embodiment, the various processing elements **1070**, **1080** may reside in the same die package.

The first processing element **1070** may further include memory controller logic (MC) **1072** and point-to-point (P-P) interfaces **1076** and **1078**. Similarly, the second processing element **1080** may include a MC **1082** and P-P interfaces **1086** and **1088**. As shown in FIG. **9**, MC's **1072** and **1082** couple the processors to respective memories, namely a memory **1032** and a memory **1034**, which may be portions of main memory locally attached to the respective processors. While the MC **1072** and **1082** is illustrated as integrated into the processing elements **1070**, **1080**, for alternative embodiments the MC logic may be discrete logic outside the processing elements **1070**, **1080** rather than integrated therein.

The first processing element **1070** and the second processing element **1080** may be coupled to an I/O subsystem **1090** via P-P interconnects **1076 1086**, respectively. As shown in FIG. **9**, the I/O subsystem **1090** includes P-P interfaces **1094** and **1098**. Furthermore, I/O subsystem **1090** includes an interface **1092** to couple I/O subsystem **1090** with a high performance graphics engine **1038**. In one embodiment, bus **1049** may be used to couple the graphics engine **1038** to the I/O subsystem **1090**. Alternately, a point-to-point interconnect may couple these components.

In turn, I/O subsystem **1090** may be coupled to a first bus **1016** via an interface **1096**. In one embodiment, the first bus **1016** may be a Peripheral Component Interconnect (PCI) bus, or a bus such as a PCI Express bus or another third generation I/O interconnect bus, although the scope of the embodiments are not so limited.

As shown in FIG. **9**, various I/O devices **1014** (e.g., biometric scanners, speakers, cameras, sensors) may be coupled to the first bus **1016**, along with a bus bridge **1018** which may couple the first bus **1016** to a second bus **1020**. In one embodiment, the second bus **1020** may be a low pin count (LPC) bus. Various devices may be coupled to the second bus **1020** including, for example, a keyboard/mouse **1012**, communication device(s) **1026**, and a data storage unit **1019** such as a disk drive or other mass storage device which may include code **1030**, in one embodiment. The illustrated code **1030** may implement one or more aspects of the process **100** (FIG. **1**), the method **600** (FIG. **2**), control of architecture **300** (FIG. **3**) and/or control of the architecture **400** (FIG. **4**) and/or the memory cells **502**-**504** (FIG. **5**), already discussed. Further, an audio I/O **1024** may be coupled to second bus **1020** and a battery **1010** may supply power to the computing system **1000**.

Note that other embodiments are contemplated. For example, instead of the point-to-point architecture of FIG. **9**, a system may implement a multi-drop bus or another such communication topology. Also, the elements of FIG. **9** may alternatively be partitioned using more or fewer integrated chips than shown in FIG. **9**.

Additional Notes and Examples

Example 1 includes a computing system comprising a system-on-chip that includes a plurality of registers, and access circuitry to access the plurality of registers, wherein the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations, and a memory including a set of executable program instructions, which when executed by the system-on-chip, cause the computing system to write data to the first portion of the plurality of registers, and transfer the data from the first portion to the second portion.

Example 2 includes the computing system of Example 1, wherein the instructions, when executed, further cause the computing system to transfer the data in response to a transfer clock pulse being identified.

Example 3 includes the computing system of Example 2, wherein the instructions, when executed, further cause the computing system to write the data to the first portion during one or more first clock cycles prior to the transfer clock pulse being identified.

Example 4 includes the computing system of Example 3, wherein the instructions, when executed, further cause the computing system to read the data in the first portion during one or more second clock cycles after the transfer clock pulse is identified.

Example 5 includes the computing system of Example 1, wherein the plurality of registers is to include a first register that is to be associated with input features associated with a neural network, a second register that is to be associated with filters associated with the neural network, and a third register that is to be associated with output features associated with the neural network.

Example 6 includes the computing system of any one of Examples 1 to 5, wherein the instructions, when executed, further cause the computing system to in each of a plurality of consecutive clock cycles, retrieve data from a memory and store the retrieved data in the first portion.

Example 7 includes the computing system of any one of Examples 1 to 5, wherein the instructions, when executed, further cause the computing system to in each of a plurality of consecutive clock cycles, read data from the second portion.

Example 8 includes the computing system of Example 1, wherein the instructions, when executed, further cause the computing system to control read and write operations to the first and second portions based on a first clock signal, and transfer data between the first and second portions based on a second clock signal that is to be different from the first clock signal.

Example 9 includes a semiconductor apparatus comprising one or more substrates, and logic coupled to the one or more substrates, wherein the logic is implemented in one or more of configurable logic or fixed-functionality logic hardware, the logic coupled to the one or more substrates to identify a plurality of registers that is associated with a system-on-chip, wherein the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations, write data to the first portion of the plurality of registers, and transfer the data from the first portion to the second portion.

Example 10 includes the apparatus of Example 9, wherein the logic coupled to the one or more substrates is to transfer the data in response to a transfer clock pulse being identified.

Example 11 includes the apparatus of Example 10, wherein the logic coupled to the one or more substrates is to write the data to the first portion during one or more first clock cycles prior to the transfer clock pulse being identified.

Example 12 includes the apparatus of Example 11, wherein the logic coupled to the one or more substrates is to read the data in the first portion during one or more second clock cycles after the transfer clock pulse is identified.

Example 13 includes the apparatus of Example 9, wherein the plurality of registers is to include a first register that is to be associated with input features associated with a neural network, a second register that is to be associated with filters associated with the neural network, and a third register that is to be associated with output features associated with the neural network.

Example 14 includes the apparatus of any one of Examples 9 to 13, wherein the logic coupled to the one or more substrates is to in each of a plurality of consecutive clock cycles, retrieve data from a memory and store the retrieved data in the first portion.

Example 15 includes the apparatus of any one of Examples 9 to 13, wherein the logic coupled to the one or more substrates is to in each of a plurality of consecutive clock cycles, read data from the second portion.

Example 16 includes the apparatus of Example 9, wherein the logic coupled to the one or more substrates is to control read and write operations to the first and second portions based on a first clock signal, and transfer data between the first and second portions based on a second clock signal that is to be different from the first clock signal.

Example 17 includes the apparatus of Example 9, wherein the logic coupled to the one or more substrates includes transistor channel regions that are positioned within the one or more substrates.

Example 18 includes a method comprising identifying a plurality of registers that are associated with a system-on-chip, wherein the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations, writing data to the first portion of the plurality of registers, and transferring the data from the first portion to the second portion.

Example 19 includes the method of Example 18, further comprising transferring the data in response to a transfer clock pulse being identified.

Example 20 includes the method of Example 19, further comprising writing the data to the first portion during one or more first clock cycles prior to the transfer clock pulse being identified.

Example 21 includes the method of Example 20, further comprising reading the data in the first portion during one or more second clock cycles after the transfer clock pulse is identified.

Example 22 includes the method of Example 18, wherein the plurality of registers includes a first register associated with input features associated with a neural network, a second register associated with filters associated with the neural network, and a third register associated with output features associated with the neural network.

Example 23 includes the method of any one of Examples 18 to 22, further comprising in each of a plurality of consecutive clock cycles, retrieving data from a memory and storing the retrieved data in the first portion.

Example 24 includes the method of any one of Examples 18 to 22, further comprising in each of a plurality of consecutive clock cycles, reading data from the second portion.

Example 25 includes the method of Example 18, further comprising controlling read and write operations to the first and second portions based on a first clock signal, and transferring data between the first and second portions based on a second clock signal that is different from the first clock signal.

Example 26 includes a semiconductor apparatus comprising means for identifying a plurality of registers that are associated with a system-on-chip, wherein the plurality of registers is to include a first portion dedicated to write operations and a second portion dedicated to read operations, means for writing data to the first portion of the plurality of registers, and means for transferring the data from the first portion to the second portion.

Example 27 includes the apparatus of Example 26, further comprising means for transferring the data in response to a transfer clock pulse being identified.

Example 28 includes the apparatus of Example 27, further comprising means for writing the data to the first portion during one or more first clock cycles prior to the transfer clock pulse being identified.

Example 29 includes the apparatus of Example 28, further comprising means for reading the data in the first portion during one or more second clock cycles after the transfer clock pulse is identified.

Example 30 includes the apparatus of Example 26, wherein the plurality of registers is to include a first register associated with input features associated with a neural network, a second register associated with filters associated with the neural network, and a third register associated with output features associated with the neural network.

Example 31 includes the apparatus of any one of Examples 26 to 30, further comprising in each of a plurality of consecutive clock cycles, means for retrieving data from a memory and storing the retrieved data in the first portion.

Example 32 includes the apparatus of any one of Examples 26 to 30, further comprising in each of a plurality of consecutive clock cycles, means for reading data from the second portion.

Example 33 includes the apparatus of Example 26, further comprising means for controlling read and write operations to the first and second portions based on a first clock signal, and means for transferring data between the first and second portions based on a second clock signal that is to be different from the first clock signal.

Thus, technology described herein may provide for an enhanced memory access method and multi-buffered system. Furthermore, technology may reduce the size, complexity and hardware of registers and memory architectures.

Embodiments are applicable for use with all types of semiconductor integrated circuit ("IC") chips. Examples of these IC chips include but are not limited to processors, controllers, chipset components, programmable logic arrays (PLAs), memory chips, network chips, systems on chip (SoCs), SSD/NAND controller ASICs, and the like. In addition, in some of the drawings, signal conductor lines are represented with lines. Some may be different, to indicate more constituent signal paths, have a number label, to indicate a number of constituent signal paths, and/or have arrows at one or more ends, to indicate primary information flow direction. This, however, should not be construed in a limiting manner. Rather, such added detail may be used in connection with one or more exemplary embodiments to facilitate easier understanding of a circuit. Any represented signal lines, whether or not having additional information, may actually comprise one or more signals that may travel in multiple directions and may be implemented with any suitable type of signal scheme, e.g., digital or analog lines implemented with differential pairs, optical fiber lines, and/or single-ended lines.

Example sizes/models/values/ranges may have been given, although embodiments are not limited to the same. As manufacturing techniques (e.g., photolithography) mature over time, it is expected that devices of smaller size could be manufactured. In addition, well known power/ground connections to IC chips and other components may or may not be shown within the figures, for simplicity of illustration and discussion, and so as not to obscure certain aspects of the embodiments. Further, arrangements may be shown in block diagram form in order to avoid obscuring embodiments, and also in view of the fact that specifics with respect to implementation of such block diagram arrangements are highly dependent upon the platform within which the embodiment is to be implemented, i.e., such specifics should be well within purview of one skilled in the art. Where specific details (e.g., circuits) are set forth in order to describe example embodiments, it should be apparent to one skilled in the art that embodiments can be practiced without, or with variation of, these specific details. The description is thus to be regarded as illustrative instead of limiting.

The term "coupled" may be used herein to refer to any type of relationship, direct or indirect, between the components in question, and may apply to electrical, mechanical, fluid, optical, electromagnetic, electromechanical or other connections. In addition, the terms "first", "second", etc. may be used herein only to facilitate discussion, and carry no particular temporal or chronological significance unless otherwise indicated.

As used in this application and in the claims, a list of items joined by the term "one or more of" may mean any combination of the listed terms. For example, the phrases "one or more of A, B or C" may mean A, B, C; A and B; A and C; B and C; or A, B and C.

Those skilled in the art will appreciate from the foregoing description that the broad techniques of the embodiments can be implemented in a variety of forms. Therefore, while the embodiments have been described in connection with particular examples thereof, the true scope of the embodiments should not be so limited since other modifications will become apparent to the skilled practitioner upon a study of the drawings, specification, and following claims.

We claim:

1. A computing system comprising:

a system-on-chip that includes a plurality of registers, and access circuitry to access the plurality of registers, wherein the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations; and

a memory including a set of executable program instructions, which when executed by the system-on-chip, cause the computing system to:

write data to the first portion of the plurality of registers as part of the write operations;

control the write operations for writing data into the first portion based on a first clock signal;

control the read operations for reading data from the second portion based on the first clock signal; and control data transfer from the first portion to the second portion based on a second clock signal, wherein the second clock signal is different from the first clock signal.

2. The computing system of claim **1**, wherein the instructions, when executed, further cause the computing system to execute the data transfer from the first portion to the second portion in response to a pulse of the second clock signal being identified.

3. The computing system of claim **2**, wherein the instructions, when executed, further cause the computing system to:

write the data to the first portion by writing the data to the first portion during one or more first close cycles of the first clock signal prior to the pulse of the second clock signal being identified.

4. The computing system of claim **3**, wherein the instructions, when executed, further cause the computing system to:

read, as part of the read operations, the data in the second portion during one or more clock cycles of the first clock signal after the pulse is identified.

5. The computing system of claim **1**, wherein the plurality of registers is to include:

a first register that is to be associated with input features associated with a neural network,

a second register that is to be associated with filters associated with the neural network, and

a third register that is to be associated with output features associated with the neural network.

6. The computing system of claim **1**, wherein the instructions, when executed, further cause the computing system to:

in each of a plurality of consecutive clock cycles of the first clock signal, retrieve further data from a storage and store the retrieved further data in the first portion.

7. The computing system of claim **1**, wherein the instructions, when executed, further cause the computing system to:

in each of a plurality of consecutive clock cycles of the first clock signal, read further data from the second portion as part of the read operations.

8. A semiconductor apparatus comprising:

one or more substrates; and

logic coupled to the one or more substrates, wherein the logic is implemented in one or more of configurable logic or fixed-functionality logic hardware, the logic coupled to the one or more substrates to:

identify a plurality of registers that is associated with a system-on-chip, wherein the plurality of registers include a first portion dedicated to write operations and a second portion dedicated to read operations;

write data to the first portion of plurality of registers as part of the write operations;

control the write operations for writing data into the first portion based on a first clock signal;

control the read operations for reading data from the second portion based on the first clock signal; and

control data transfer from the first portion to the second portion based on a second clock signal, wherein the second clock signal is different from the first clock signal.

9. The apparatus of claim **8**, wherein the logic coupled to the one or more substrates is further to:

execute the data transfer from the first portion to the second portion in response to a pulse of the second clock signal being identified.

10. The apparatus of claim **9**, wherein the logic coupled to the one or more substrates is to:

write the data to the first portion by writing the data to the first portion during one or more first close cycles of the first clock signal prior to the pulse of the second clock signal being identified.

11. The apparatus of claim **10**, wherein the logic coupled to the one or more substrates is to:

read, as part of the read operations, the data in the second portion during one or more clock cycles of the first clock signal after the pulse is identified.

12. The apparatus of claim **8**, where in the plurality of registers is to include:

a first register that is to be associated with the input features associated with a neural network,

a second register that is to be associated with filters associated with the neural network, and

a third register that is to be associated with output features associated with the neural network.

13. The apparatus of claim **8**, wherein the logic coupled to the one or more substrates is to:

in each of a plurality of consecutive clock cycles of the first clock signal, retrieve further data from a storage and store the retrieved further data in the first place.

14. The apparatus of claim **8**, wherein the logic coupled to the one or more substrates is to:

in each of a plurality of consecutive clock cycles of the first clock signal, read further data from the second portion as part of the read operations.

15. The apparatus of claim **8**, wherein the logic coupled to the one or more substrates includes transistor channel regions that are positioned within one or more substrates.

16. A method comprising:

identifying a plurality of registers that are associated with a system-on-chip, wherein the plurality of registers includes a first portion dedicated to write operations and a second portion dedicated to read operations;

writing data to the first portion of the plurality of registers as part of the write operations;

controlling the write operations for writing data into the first portion based on a first clock signal;

controlling the read operations for reading data from the second portion based on the first clock signal; and

controlling data transfer from the first portion to the second portion based on a second clock signal, wherein the second clock signal is different from the first clock signal.

17. The method of claim **16**, further comprising:

executing the data transfer from the first portion to the second portion in a response to a pulse of the second clock signal being identified.

18. The method of claim **17**, wherein writing the data to the first portion comprises writing the data to the first portion during one or more first clock cycles of the first clock signal prior to the pulse of the second clock signal being identified.

19. The method of claim **18**, further comprising:

reading, as part of the read operations, the data in the second portion during one or more clock cycles of the first clock signal after the pulse is identified.

20. The method of claim **16**, wherein the plurality of registers includes:

a first register associated with input features associated with a neural network,

a second register associated with filters associated with a neural network, and

a third register associated with output features associated with the neural network.

21. The method of claim **16**, further comprising:

in each of a plurality of consecutive clock cycles of the first clock signal, retrieving further data from a storage and storing is retrieved further data in the first portion.

22. The method of claim **16**, further comprising:

in each of a plurality of consecutive clock cycles of the first clock signal, reading further data from the second portion as part of the read operations.

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